

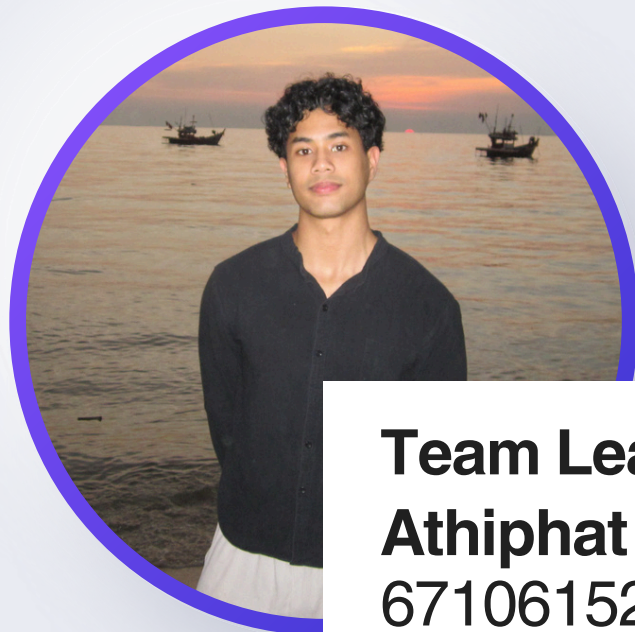
ทีม นักชุดทอง

Gold Signal Intelligence via LLM Decision & ReAct Reasoning

Data Science for Signal Processing

CN240

Our team

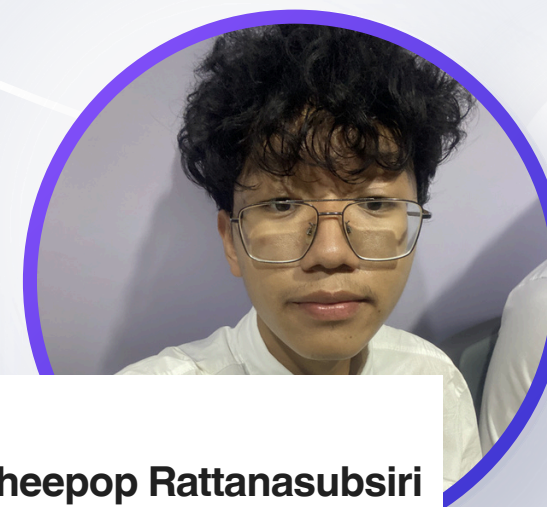


Team Lead
Athiphat Sunsit
6710615292

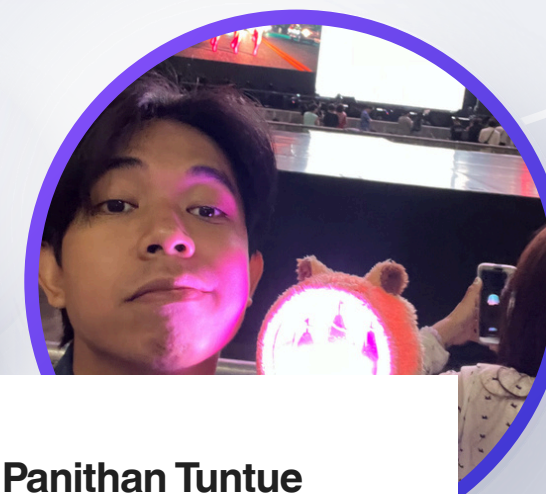
Purich Ampawa



Napattira Loaklehung
6710545010



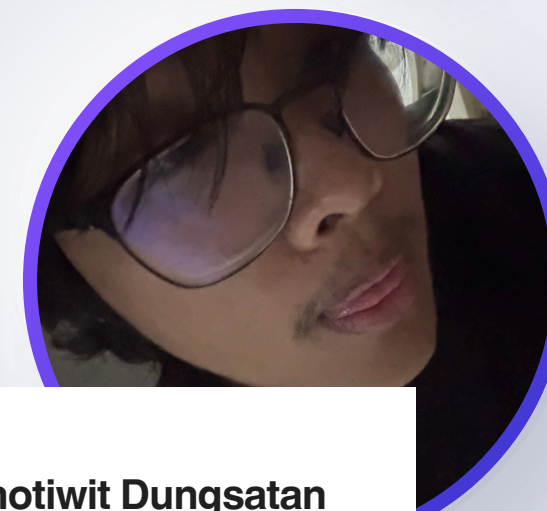
Theepop Rattanasubsiri
6710685014



Panithan Tuntue
6710615144



6710615185



Chotiwit Dungsatan
6710615060



Sitthipong Kamngam
6710615284



Phatcharaphon Malaisri
6710685055

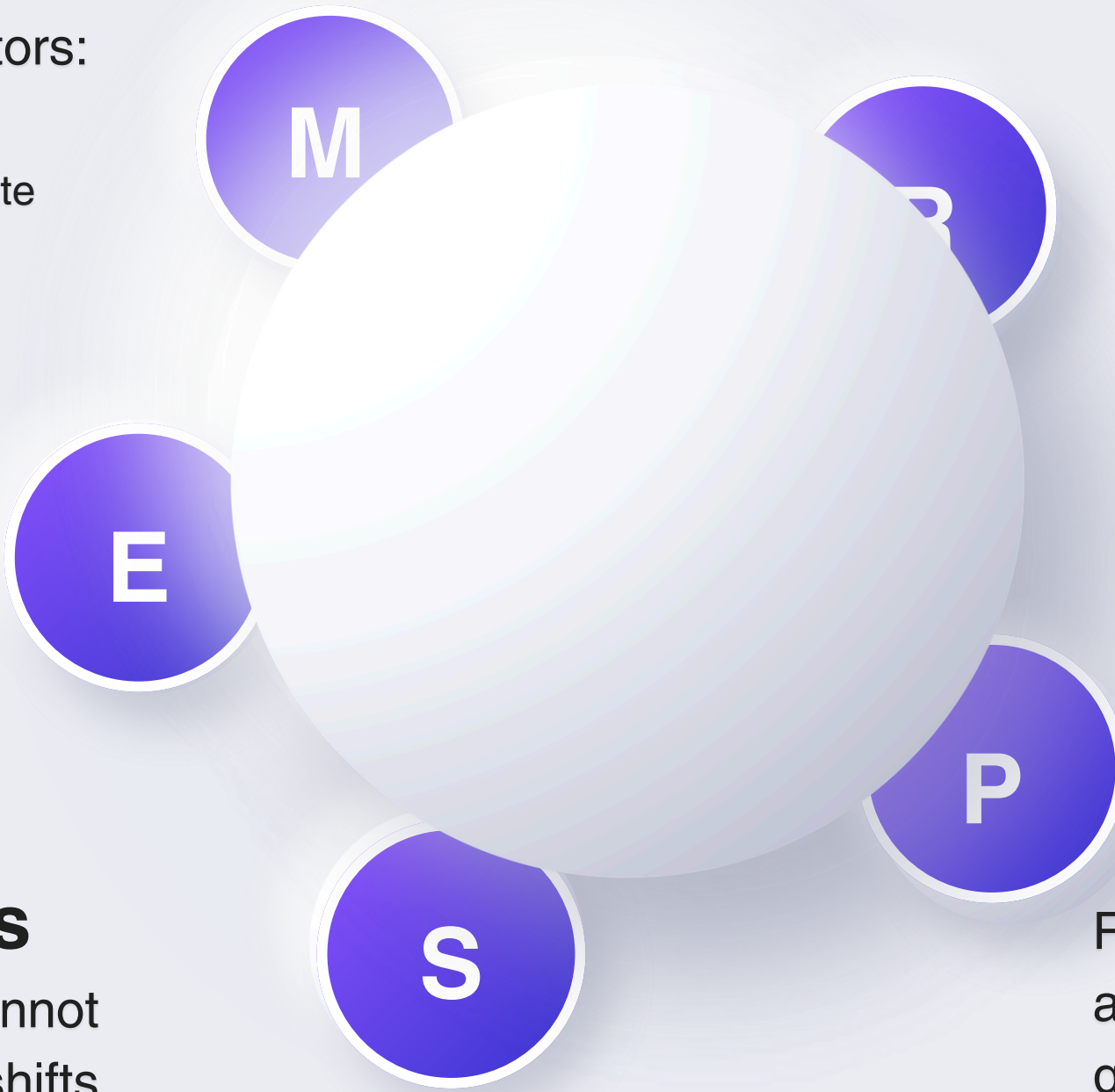


Lalita Thatsananunchai
6710615243



Benchaphon Pinakasa
6710625028

PROBLEM STATEMENT



Market Complexity

Gold prices are driven by factors:

- geopolitical events
- Fed interest rate policy
- inflation data
- USD/THB exchange rate
- market sentiment

Extreme Volatility

A single news headline can move prices by 1–2% within minutes

System Limitations

Traditional rule-based systems cannot adapt to sudden market shifts

Retail Financial Risk

Retail traders on Aom NOW must decide BUY/SELL/HOLD with minimum ฿1,000 per transaction — a wrong decision has real financial consequences

Proposed Solution / Research Goal

Research Question: Can a multi-source LLM agent outperform simple technical analysis in generating reliable gold trading signals

“WHY USE LLM ?

NOT RULE-BASED”

Rule-based systems

have hard limits:

fixed logic like "*RSI > 70 = SELL*"
cannot read news context

GOAL

Use an LLM as an "**analyst**" that receives
market data & makes reasoned decisions

LLM systems

can do more:

- Synthesize multiple data sources simultaneously
(**price + indicators + news + portfolio**)
- Produce human-readable reasoning
- Behavior can be adjusted via System Prompt
without touching code

Mission & System Overview



Analyze the gold market & Generate portfolio-aware trading signals



BUY / SELL / HOLD signal with confidence score and rationale



Target platform: **Aom NOW (Hua Seng Heng)**
minimum \$1,000 per order, trades in grams

Starting capital: \$1,500



Core stack: **Python +**
Gradio Dashboard +
PostgreSQL + LLM API
(Gemini / Groq / Claude)

System Architecture

UI Layer

Gradio dashboard with 4 tabs:

- Live Analysis
- Run History
- Portfolio
- Trace Log

Data Engine

- GoldDataFetcher (yfinance)
- TechnicalIndicators
- GoldNewsFetcher (RSS + FinBERT sentiment)

Agent Orchestration

ReAct loop:

- PromptBuilder constructs the context
- LLMClient calls the model
- ReactOrchestrator manages the reasoning loop

Persistence

- PostgreSQL stores every run
- signal
- market snapshot
- portfolio state

Block-level Architecture **DIAGRAM**

Production Environment

live data layer

yfinance → GoldDataFetcher
RSS/News → GoldNewsFetcher
→ TechnicalIndicators(RSI, MACD, EMA, ATR)

Merge Portfolio

Fetches from DB via service

BackTest Environment

Mocked data layer

Price CSV → Pandas Resampler
News CSV → HistoricalNewsLoader
→ _ensure_indicators() (RSI, MACD, EMA, ATR)

Merge SimPortfolio

Fetches from memory

full_market_state dict

AI agent core

ReAct: Thought → Action → Observation → FINAL_DECISION(LLM call)

full_decision dict

Multi-interval voting

1h (22%), 4h (30%), 1d(12%)
calculate_vote() → voting_result

Persistence & UI layer

PostgreSQL (RunDB) • Gradio Dashboard

Simulated execution

Loop per candle:
CandleCache → _apply_to_portfolio()

Metrics & I/O layer

Win rate • PnL • Shape → CSV export

DATA SOURCES & FEATURE ENGINEERING

Raw data

XAU/USD, USD/THB,
Thai gold price, latest
5 OHLCV candles

Price source

yfinance
Twelvedata

Exchange rate

Real-time
USD/THB
from intergold

Thai gold price

Extract gold
96.5% price from
intergold

Features used

Momentum : RSI-14, MACD,
Trend: EMA (20, 50), Crossover
Signals
Volatility: Bollinger Bands, ATR-14

All data is serialized
into a market_state
dict before being
passed to the AI
Agent

DATA & FEATURES 3

CATEGORIES OF INPUT FEATURES

PRICE DATA

- **fetch OHLCV history**
via yfinance(GC=F ticker)
- **Live spot price in USD/oz**
- **Live spot price in THB/baht**
- **Live USDTHB exchange rate**

TECHNICAL INDICATORS

- **RSI (14-period)**
overbought/oversold signal
- **MACD (12,26,9)**
momentum & trend direction
- **EMA 20 / EMA 50**
uptrend / downtrend / neutral
- **Bollinger Bands**
volatility range

NEWS & SENTIMENT

- **8 news categories**
gold_price, usd_thb, fed_policy, inflation, geopolitics, dollar_index, thai_economy, thai_gold_market
- **Sources**
RSS Kitco, Reuters, Bangkok Post+ yfinance metadata
- **FinBERT scoring**
Hugging Face API, no GPU needed
- **Token-budget packing**
greedy selection of top articles

Output File → “market_state.json”

```
"meta": {
  "agent": "gold-trading-agent",
  "version": "1.1.0",
  "generated_at": "2026-03-31T10:25:57.018291+07:00",
  "history_days": 30,
  "interval": "1h"
},
"data_quality": {
  "quality_score": "good",
  "is_weekend": false,
  "llm_instruction": "Use standard technical analysis.",
  "warnings": []
},
"data_sources": {
  "price": "twelvedata",
  "usd_thb_rate": "intergold_live_stream",
  "thai_gold": "intergold_live_stream",
  "news": "newsfetcher"
},
"market_data": {
  "spot_price_usd": {
    "source": "twelvedata",
    "price_usd_per_oz": 4562.0438,
    "timestamp": "2026-03-31T10:25:39.859259+07:00",
    "confidence": 0.916
  },
  "usd_thb_rate": {
    "source": "intergold_live_stream",
    "usd_thb": 32.84,
    "timestamp": "2026-03-31T10:24:41.973"
  },
  "thai_gold_thb": {
    "source": "intergold_live_stream",
    "sell_price_thb": 70942.0,
    "buy_price_thb": 70847.0,
    "gold_spot_usd": 4561.04,
    "usd_thb_live": 32.84,
    "timestamp": "2026-03-31T10:24:41.973"
  },

```

```
"recent_price_action": [
  {
    "datetime": "2026-03-31T06:00:00+07:00",
    "open": 4539.10009765625,
    "high": 4550.10009765625,
    "low": 4520.7998046875,
    "close": 4542.10009765625,
    "volume": 2139
  },
  {
    "datetime": "2026-03-31T07:00:00+07:00",
    "open": 4541.39990234375,
    "high": 4577.2998046875,
    "low": 4510.0,
    "close": 4576.60009765625,
    "volume": 7357
  },
  {
    "datetime": "2026-03-31T08:00:00+07:00",
    "open": 4576.5,
    "high": 4649.5,
    "low": 4560.89990234375,
    "close": 4635.5,
    "volume": 14017
  },
  {
    "datetime": "2026-03-31T09:00:00+07:00",
    "open": 4634.7001953125,
    "high": 4637.60009765625,
    "low": 4595.7001953125,
    "close": 4597.7998046875,
    "volume": 5594
  },
  {
    "datetime": "2026-03-31T10:00:00+07:00",
    "open": 4597.5,
    "high": 4607.2998046875,
    "low": 4596.0,
    "close": 4600.2001953125,
    "volume": 1017
  },

```

```
"technical_indicators": {
  "rsi": {
    "value": 59.93,
    "signal": "neutral",
    "period": 14
  },
  "macd": {
    "macd_line": 16.4579,
    "signal_line": 12.5688,
    "histogram": 3.8891,
    "crossover": "none"
  },
  "bollinger": {
    "upper": 4619.6,
    "middle": 4567.66,
    "lower": 4515.72,
    "bandwidth": 0.022744,
    "pct_b": 0.8132,
    "signal": "inside"
  },
  "atr": {
    "value": 31.67,
    "period": 14,
    "volatility_level": "high"
  },
  "trend": {
    "ema_20": 4564.6,
    "ema_50": 4540.52,
    "trend": "uptrend",
    "golden_cross": false,
    "death_cross": false
  },
  "latest_close": 4600.2,
  "calculated_at": "2026-03-31T10:25:40"
}
```



```
{
  "cash_balance":      1500.00,    # ฿
  "gold_grams":        0.0,        # g
  "cost_basis_thb":    0.00,       # ฿
  "current_value_thb": 0.00,       # ฿
  "unrealized_pnl":    0.00,       # ฿
  "trades_today":      0,
  "updated_at":        "2026-03-29T10:00:00Z"
}
```

portfolio.json

Output File → “market_state.json”

```
"by_category": {
  "dollar_index": {
    "label": "ดัชนีค่าเงินดอลลาร์ (DXY)",
    "impact": "medium",
    "tickers": [
      "DX-Y.NYB",
      "UUP"
    ],
    "count": 5,
    "articles": [
      {
        "title": "War sets dollar for monthly rise, yen recovers on intervention threat",
        "url": "https://sg.finance.yahoo.com/news/war-sets-dollar-monthly-rise-004541236.html",
        "source": "Reuters",
        "published_at": "2026-03-31T07:45:41+07:00",
        "ticker": "DX-Y.NYB",
        "category": "dollar_index",
        "impact_level": "medium",
        "sentiment_score": 0.6551
      },
      {
        "title": "Gold Edges Lower Amid Divergent Signals on U.S.-Iran Talks",
        "url": "https://www.wsj.com/finance/commodities-futures/gold-edges-lower-amid-divergent-signals-on-u-s-iran-talks-9bd4841b?siteid=yhoof2&yptr=yahoo",
        "source": "The Wall Street Journal",
        "published_at": "2026-03-31T06:43:00+07:00",
        "ticker": "DX-Y.NYB",
        "category": "dollar_index",
        "impact_level": "medium",
        "sentiment_score": 0.6551
      },
      {
        "title": "Famous short seller targets crypto-friendly bank in new report",
        "url": "https://www.thestreet.com/crypto/markets/famous-short-seller-targets-crypto-friendly-bank-in-new-report",
        "source": "TheStreet",
        "published_at": "2026-03-31T06:28:44+07:00",
        "ticker": "DX-Y.NYB",
        "category": "dollar_index",
        "impact_level": "medium",
        "sentiment_score": -0.9006
      }
    ]
  }
}
```

Portfolio-Aware Constraints (enforced in system prompt)

can_buy

= YES only if cash
balance $\geq$ ฿1,000

can_sell

= YES only if gold
holdings $>$ 0
grams

- HOLD IS ONLY VALID WHEN BOTH
CAN_BUY AND CAN_SELL ARE NO

- ALL PRICES OUTPUT IN THB, NOT USD

TEAM AI AGENT

PromptBuilder & Context Injection

JSON PAYLOAD CONSTRUCTION

The PromptBuilder takes processed data from the Data Engine and converts it into a structured JSON format.

CONTEXTUAL AWARENESS

Injects real-time parameters: Current Price, Technical Signals, and Portfolio Status (\$1,500 capital).

MARKET STATE

JSON PAYLOAD

LLM PROMPT

the flow

Input to LLM Before ReAct Loop

```
PromptPackage(  
  SYSTEM:  
    You are a Gold Market Analyst.  
    You MUST output a final trading decision as  
    a single JSON object (no markdown fences).  
    Fields: action (FINAL_DECISION), signal (BUY/SELL/HOLD),  
    confidence (0-1), entry_price, stop_loss, take_profit, rationale.  
  
  USER:  
    ### MARKET STATE  
    Gold: $4495.0 | RSI(14): 57.68 [neutral]  
    MACD: 25.8452/17.7424 hist:8.1027  
    Trend: EMA20=4490.2 EMA50=4473.78 [uptrend]  
    News Highlights:  
  
    — Portfolio —  
    Cash:          $1,500.00  
    Gold:          0.0000 g  
    Cost basis: $0.00  
    Cur. value: $0.00  
    Unreal PnL: $0.00  
    Trades today: 0  
    can_buy:  YES  
    can_sell: NO (gold_grams = 0)  
    — End Portfolio —
```

json file

TEAM AI AGENT

DesignReAct Loop

01

THOUGHT

Market state +
portfolio context
loaded into a
structured
prompt

02

REASONING

LLM analyzes RSI,
MACD, trend,
news sentiment,
and trading
constraints

03

DECISION

Outputs a single
JSON: signal
(BUY/SELL/HOLD),
confidence (0-1),
entry price, stop
loss, take profit,
and rationale

Output After LLM THOUGHT_FINAL

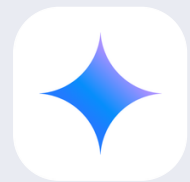
```
{  
  "action": "FINAL_DECISION",  
  "signal": "SELL",  
  "confidence": 0.75,  
  "entry_price": 70010,  
  "stop_loss": 69990,  
  "take_profit": 70210.00,  
  "rationale": "Gold is showing an extremely overbought condition with RSI(14) at 100.0,  
    indicating a high probability of a near-term pullback or consolidation,  
    despite the prevailing uptrend (EMA20 > EMA50).  
    The current price is significantly extended above both EMAs,  
    suggesting it's vulnerable to a correction. Furthermore,  
    the US Dollar is strengthening, which typically exerts bearish pressure on gold prices.  
    While falling treasury yields could be supportive,  
    the focus on upcoming Fed data introduces uncertainty.  
    The dominant signal is the unsustainable overbought RSI,  
    making a short-term SELL advisable to capitalize on an expected correction."  
}
```

json file

LLM Providers & Fault Tolerance

LLM Providers

- Supports 6 providers via **Factory Pattern**



Gemini



Groq



Ollama

- Adding a new provider takes 3 lines
no changes to business logic

Fault Tolerance

- Retries up to 3 times with exponential backoff (1s → 2s → 4s)
- JSON parse fallback
 - LLM responds with non-JSON
 - defaults to signal=HOLD, confidence=0.0
- ValueError (invalid input)
 - stops immediately
 - no retry since retrying won't help

Singleton Portfolio Pattern & UPSERT

- Portfolio uses exactly 1 row always (id=1) in PostgreSQL

- Saved with UPSERT:

INSERT ... ON CONFLICT (id) DO UPDATE SET ...

- Benefits:

No duplicate state, simple to query, always consistent

- Schema covers:

cash_balance, gold_grams, cost_basis_thb,
current_value_thb, unrealized_pnl, trades_today

- Run history stores every run with full react_trace and market_snapshot (as JSON)

RiskManager & Trading Constraints

Hard Constraints (enforced via System Prompt):

- $\text{Cash} < \text{฿}1,000 \rightarrow \text{can_buy} = \text{NO}$ (Aom NOW minimum order is ฿1,000)
- $\text{gold_grams} = 0 \rightarrow \text{can_sell} = \text{NO}$ (no short selling)
- Maximum risk per trade: 50% of capital
- entry/stop_loss/take_profit must be in THB, not USD

Soft Constraints (LLM reasoning):

- $\text{RSI} > 70 = \text{overbought} \rightarrow \text{avoid BUY}$
- $\text{RSI} < 30 = \text{oversold} \rightarrow \text{avoid SELL}$
- Risk/Reward: $\text{take_profit} - \text{entry}$ must always exceed $\text{entry} - \text{stop_loss}$

RESULT & PERFORMANCE

Average tokens per run
~650 tokens

End-to-end latency
(**Gemini Flash**) < 5 seconds

LLM providers supported = 6

Articles **analyzed** per run Up
to 30 (token-packed)

News categories
tracked = 8

Token **reduction vs.**
baseline ~40%

Evaluation Metrics (Actually Used)



Directional Accuracy

% of signals where predicted price direction is correct



Avg Net PnL per trade

Average profit/loss per executed signal



Signal Sensitivity

Ratio of BUY+SELL vs HOLD (is the system active or over-conservative?)



Targets

Directional Accuracy > 55%,
Signal Sensitivity > 40%

Results measured separately by provider (Gemini vs Groq) and timeframe combination

Financial Risk Metrics



Sharpe Ratio:

- Return relative to total risk (target > 1.0)



Sortino Ratio:

- Like Sharpe but penalizes only downside deviation



Max Drawdown:

- Largest peak-to-trough loss (target < 20%)



Win Rate:

- % of trades that are profitable (target > 50%)



Profit Factor:

- Gross profit / gross loss (target > 1.5)

- All metrics compared against a Buy & Hold benchmark

Try Backtest & Result

engine : **SimPortfolio** Data : **90 days** candles : **1 hr**

Backtest Analysis: Price & Signals



PORTFOLIO SUMMARY

Initial Capital	1,500 THB
Final Portfolio	972.00 THB
Total Return	-35.20%

RISK METRICS

Max Drawdown	-35.20%
Sharpe Ratio	-8.05
Sortino Ratio	-8.01

STRATEGY STATS

Win Rate	25.0%
Directional Acc.	25.00%
Total Signals	16
BUY / SELL	16 / 0
Avg Confidence	0.644

Risks & Mitigation

Price mismatch (USD vs THB)

Impact:

Wrong entry price.

Mitigation:

Force THB output in system prompt

HOLD bias

Impact:

System too inactive.

Mitigation:

0.70 threshold + signal distribution monitoring

Look-ahead bias in backtest

Impact:

Overfitted results.

Mitigation:

Use only data available at that point in time

RAM / latency (FinBERT API)

Impact:

Slow response.

Mitigation:

3,000 token budget + Greedy Packing



นักขุดทอง

Error Handling & Reliability

FinBERT API

- retry ×3, 429 rate limit → sleep 10s,
- 503 cold start → sleep estimated_time

LLM calls

retry ×3 with 2-second delay

JSON parsing

fallback to HOLD on malformed output

News fetch

- per-category error logging
- system continues even if one category fails

Database

UPSERT pattern prevents duplicate portfolio rows

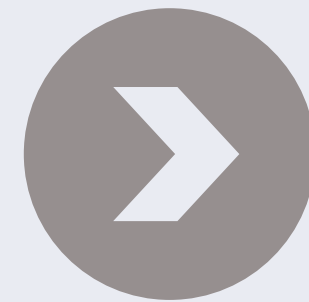
Real Challenges Encountered & Solutions

● Rate limiting

Rate limiting



LLM API slow or throttled



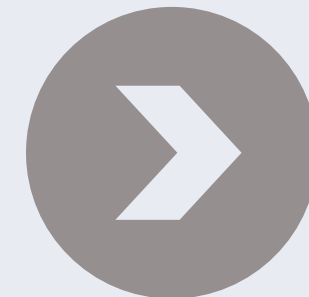
exponential backoff + Mock provider for development to ensure pipeline continuity without manual resets.

● JSON parse failures

JSON parse failures



LLM returns markdown or free text instead of JSON breaking the execution logic.

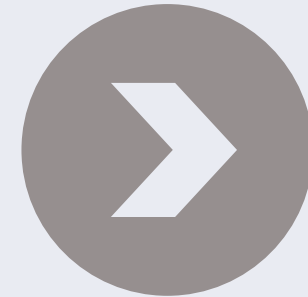


defaults to HOLD →
extract_json() strips fences
+ regex fallback

Real Challenges Encountered & Solutions

Gemini vs Groq confidence gap

Gemini vs Groq
confidence gap



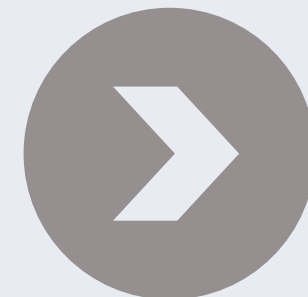
Gemini consistently
returns higher confidence
than Groq
biasing the weighted average.



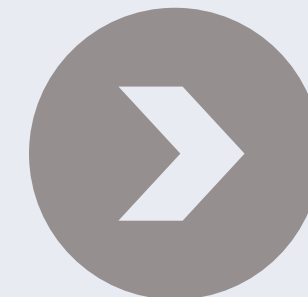
normalize confidence per
provider, or rely on weighted
voting to reduce bias

FinBERT cold start

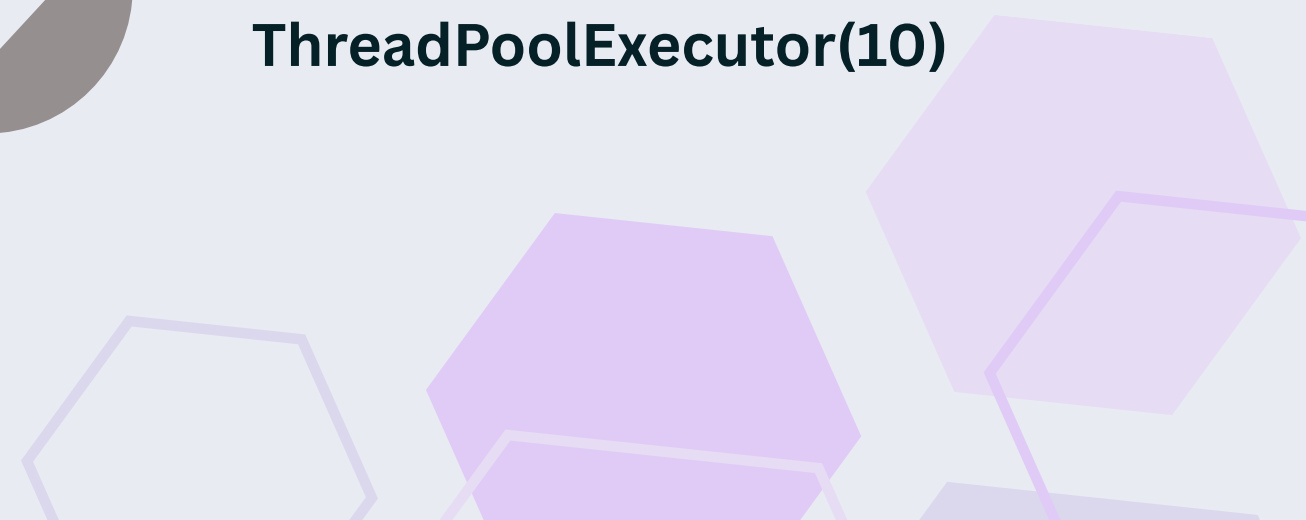
FinBERT cold start



Inference API slow on
first call (lowed down
real-time analysis.)



parallel fetching with
`ThreadPoolExecutor(10)`

Several overlapping purple hexagons of varying shades in the bottom right corner of the slide.

Summary & Future Work



Production-grade ReAct agent

Integrating multiple data sources. Combines live prices, technical indicators, and news sentiment into a single traceable trading decision.



Multi-interval Weighted Voting

Runs LLM analysis across 1h, 4h, and 1d — final signal determined by weighted confidence score, to reduce single-timeframe noise



Portfolio-aware Constraints

Injects cash and gold holdings into every prompt, automatically enforcing minimum buy and no short-selling rules, to preventing invalid trades



Pluggable LLM Factory

Supports Gemini, Groq, and Mock behind a unified interface. Switching providers requires only three lines of code.

SUMMARY

FUTURE WORK

Enterprise data ☐

TwelveData / Bloomberg for real-time tick data

Cloud deployment ☐

Render / GCP for production use

Auto-execution ☐

Direct integration with Aom NOW API (if available)

Reinforcement Learning ☐

Automatically tune interval weights from backtest results